Lecture 27

The scalar Riemann–Hilbert problem

Objectives. *A prescribed boundary jump determines an analytic function on each side of a contour once a normalisation removes the entire-function ambiguity. We solve an additive problem on the real line and then one rational Wiener–Hopf problem. The upper and lower half-plane conventions determine every factor and every time-support sign.*

By the end of this lecture you will be able to:

- Solve a compact smooth additive jump by a normalised Cauchy integral.
- Prove uniqueness by gluing and Liouville's theorem.
- Factor the chosen rational coefficient and solve its boundary equation.
- Convert the upper and lower solution components to one-sided time functions.

27.1 Boundary values and gluing

Orient the real axis from left to right. Its left side is the upper half-plane, denoted by $+$; the lower half-plane is denoted by $-$. A *scalar Riemann–Hilbert problem* prescribes a relation between the boundary values of scalar holomorphic functions on the two sides. Here boundary values are continuous extensions from each closed half-plane. The normalisation $O(1/z)$ means a bound $C/|z|$ uniform in that half-plane for sufficiently large $|z|$.

Lemma 27.1 (Gluing)**.** *Suppose F_+ and F_- are holomorphic on the upper and lower half-planes, continuous up to the real line, and agree there. Their union is entire.*

Proof. The union is continuous. Take a triangle crossing the real line, initially with no vertex on that line and no edge contained in it. Cut its upper portion along height ε and its lower portion along height $-\varepsilon$. Each resulting polygon is contained in a half-plane, so its boundary integral is zero by triangulation and the Cauchy theorem, Theorem 9.4. As $\varepsilon \downarrow 0$, their exterior edges tend to the original triangle boundary. The omitted edge segments have length $O(\varepsilon)$ and bounded integrand, so their integrals tend to zero. The horizontal cut integrals have opposite orientations. Continuity on a compact neighbourhood makes them converge to the same real-line integral, so they cancel. The triangle integral of the union is therefore zero. Triangles in the exceptional positions are limits of these triangles; continuity again passes their integrals to the limit. Morera's theorem, Theorem 10.8, now proves holomorphy everywhere. ∎

If both functions also satisfy the stated $O(1/z)$ bound, the glued entire function tends to zero at infinity. It is bounded on the remaining compact disk, so Liouville's theorem, Theorem 10.4, makes it identically zero. This will be the uniqueness argument for both problems.

27.2 An additive jump on the real line

Let $h \in C_c^1(\mathbb{R})$, meaning a continuously differentiable function with compact support. Define the normalised Cauchy integral by

$$C_h(z) = \frac{1}{2\pi \mathrm{i}} \int_{\mathbb{R}} \frac{h(t)}{t - z}\, \mathrm{d}t, \qquad z \notin \mathbb{R}. \tag{27.1}$$

The normalising factor is part of the definition. With $\rho = h$ in Equation (12.2), the exact relation is $C_h(z) = F(z)/(2\pi\mathrm{i})$. Thus the jump $2\pi\mathrm{i}\rho$ from Lecture 12 becomes the jump h here. Differentiation under the integral on compact subsets of either half-plane proves holomorphy.

Theorem 27.2 (Additive problem). *There is a unique pair F_+, F_-, holomorphic in their respective half-planes, continuous up to the real line and uniformly $O(1/z)$ at infinity, such that $F_+(x) - F_-(x) = h(x)$ for every $x \in \mathbb{R}$. It is the restriction of C_h to those half-planes, with boundary values*

$$C_h^{\pm}(x) = \frac{1}{2\pi \mathrm{i}}\, \mathrm{PV} \int_{\mathbb{R}} \frac{h(t)}{t - x}\, \mathrm{d}t \;\pm \frac{1}{2} h(x). \tag{27.2}$$

Proof. For fixed x, put $s = t - x$. The Plemelj lemma, Theorem 8.2, applies to $h(x + s)$ and the denominator $s \mp \mathrm{i}y$. After division by $2\pi\mathrm{i}$ it gives the displayed boundary values. We also check the continuity required in the statement.

On $|s| < 1$, subtract $h(x)$ from $h(x + s)$. The principal-value contribution of the constant to the real kernel is zero. The quotient $(h(x + s) - h(x))/s$ is bounded by $\|h'\|_\infty$. Dominated convergence on this interval and on $|s| \geq 1$ shows that the principal value depends continuously on x, locally uniformly in x. For a full approach from a half-plane, write $(s - \mathrm{i}y)^{-1} = s/(s^2 + y^2) + \mathrm{i}y/(s^2 + y^2)$. After the same subtraction, the error of the real kernel on $|s| < 1$ is bounded in integral by $\|h'\|_\infty \int_{-1}^{1} y^2/(s^2 + y^2)\, \mathrm{d}s = O(|y|)$. The imaginary remainder is bounded by $\|h'\|_\infty \int_{-1}^{1} |ys|/(s^2 + y^2)\, \mathrm{d}s = O(|y| \log(1/|y|))$. The constant part tends to $\mathrm{i}\pi\, \mathrm{sgn}(y) h(x)$. The tails converge uniformly for x in a compact set, since h has compact support and their denominators stay away from zero. These bounds prove continuous extension from both half-planes.

If $\operatorname{supp} h \subset [-R, R]$ and $|z| > 2R$, then $|t - z| \geq |z|/2$ on the support. Thus $|C_h(z)| \leq \|h\|_1 / (\pi |z|)$, uniformly on both sides. The two boundary values differ by h, proving existence. For two solutions, subtract them. Their boundary values now agree, and the gluing lemma followed by Liouville proves their difference zero. ∎

The jump is independent of the principal-value term because that term is the same on both sides. Reversing the real-axis orientation would change which side is called + and would change this convention.

27.3 A rational zero-index coefficient

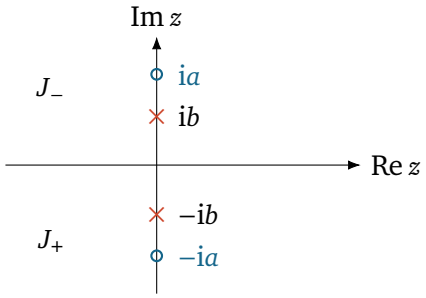
For a continuous nonvanishing real-axis function tending to 1 at both ends, its *index* is the total change of a continuous argument divided by 2π, following the axis from $-\infty$ to $+\infty$. The endpoints are identified by that common value, so the change is an integer multiple of 2π.

Fix $a, b > 0$ and choose the coefficient

$$J(z) = \frac{z^2 + a^2}{z^2 + b^2} = J_+(z)J_-(z), \qquad J_+(z) = \frac{z + \mathrm{i}a}{z + \mathrm{i}b}, \quad J_-(z) = \frac{z - \mathrm{i}a}{z - \mathrm{i}b}. \tag{27.3}$$

This is a product factorisation, rather than a quotient convention. For real $z = x$, the coefficient is positive. Its argument is identically zero, so its index is zero. Each factor is analytic and zero-free in its named half-plane and equals $1 + O(1/z)$ there. The reciprocal of each factor has the same properties.

The zeros and poles determine their half-plane allocation.



Circles denote zeros and crosses denote poles of the factors. The labels J_- and J_+ identify the upper and lower zero/pole pairs, respectively. The diagram illustrates $a > b$; the formulas allow any $a, b > 0$, including $a = b$, when the corresponding zero and pole cancel. The factor J_+ places both of its exceptional points below its domain of analyticity.

27.4 The rational boundary equation

A *Wiener–Hopf boundary equation* here asks for functions analytic in opposite half-planes whose boundary values obey a given scalar equation. For the fixed coefficient above, solve

$$J(x)U_+(x) + U_-(x) = \frac{1}{x^2 + b^2}, \qquad x \in \mathbb{R}, \tag{27.4}$$

with continuous boundary values and $U_\pm(z) = O(1/z)$ uniformly in their respective half-planes.

The factorisation must put the unknowns into separate analytic classes. After division by the lower factor, the term containing U_+ has only upper-half-plane factors, and the term containing U_- has only lower-half-plane factors. The transformed forcing must then be separated in the same way. Partial fractions can perform that separation because their poles identify which half-plane each term is allowed to occupy.

Divide by J_- and split the resulting rational forcing:

$$\frac{1}{(z^2 + b^2)J_-(z)} = \frac{1}{(z + \mathrm{i}b)(z - \mathrm{i}a)} = \frac{\mathrm{i}}{a + b}\frac{1}{z + \mathrm{i}b} - \frac{\mathrm{i}}{a + b}\frac{1}{z - \mathrm{i}a}.$$

Call the first term $G_+(z)$ and the second $G_-(z)$. Each is analytic and $O(1/z)$ in its named half-plane. Thus the equation is solved by $J_+U_+ = G_+$ and $U_-/J_- = G_-$, giving

$$U_+(z) = \frac{\mathrm{i}}{(a + b)(z + \mathrm{i}a)}, \qquad U_-(z) = -\frac{\mathrm{i}}{(a + b)(z - \mathrm{i}b)}. \tag{27.5}$$

Multiplication by J_+ and division by J_- recover the split terms. Thus these functions satisfy the divided equation and hence the original equation and its boundary conditions. For example, $a = 2$ and $b = 1$ give $U_+(0) = 1/6$.

The reconstruction reverses the preparation: divide the upper split term by the upper factor, and multiply the lower split term by the lower factor. Check analyticity and decay of the resulting functions, then substitute them into the original boundary equation. Checking only the sum of the split terms would verify the transformed forcing but would not yet verify the original unknowns.

For uniqueness, subtract two solutions. The upper function $J_+\Delta U_+$ and the lower function $-\Delta U_-/J_-$ agree on the line. Both decay uniformly as $O(1/z)$. Gluing and Liouville make them zero. The factors have no zeros in their respective domains, so the original differences vanish. This proves uniqueness for the specified problem.

Define the time transform by $\mathcal{T}f(\nu) = \int_{\mathbb{R}} f(t)e^{\mathrm{i}\nu t}\,\mathrm{d}t$. It equals $\sqrt{2\pi}\,\widehat{f}(-\nu)$ in the unitary spatial convention. Direct exponential integration gives

$$\mathcal{T}\left(\frac{e^{-at}\mathbb{1}_{(0,\infty)}(t)}{a+b}\right) = U_+(\nu), \qquad \mathcal{T}\left(\frac{e^{bt}\mathbb{1}_{(-\infty,0)}(t)}{a+b}\right) = U_-(\nu).$$

The first integral is $1/((a+b)(a-\mathrm{i}\nu))$. The second is $1/((a+b)(b+\mathrm{i}\nu))$. Their poles are respectively below and above the real axis. This explicitly verifies the connection between support and half-plane analyticity used in Lecture 26; no sign is inferred from a drawing alone.

27.5 What a nonzero index changes

The general existence theory is a cited extension, not a consequence of the one rational calculation. A precise version from Ablowitz–Fokas, §7.3.1, pp. 529–531, concerns a counterclockwise unit circle C. Assume its coefficient J_C is nonvanishing and Hölder continuous, and its forcing f_C is Hölder continuous. Hölder continuity means a bound $|f_C(t)-f_C(s)| \le M|t-s|^\beta$ for some M and $0 < \beta < 1$. The circle index $\operatorname{ind}_C J_C$ is the total change of a continuous argument of $J_C(e^{\mathrm{i}t})$ as t runs from 0 to 2π, divided by 2π. The argument is followed continuously along the full circuit, including any full turns. The inside is now the + side. Seek continuous boundary values with $\Phi_+ = J_C\Phi_- + f_C$ and exterior normalisation $\Phi_-(z) = O(1/z)$.

The cited factor construction gives analytic factors X_+ inside and X_- outside. They have no zeros at finite points of their respective domains. They satisfy $X_+ = J_C X_-$ and $X_-(z) \sim z^{-\operatorname{ind}_C J_C}$ at infinity. For index zero there is a unique solution for every such forcing. For positive index, solutions exist for every forcing and the homogeneous solution space has dimension $\operatorname{ind}_C J_C$. For negative index, existence requires the moment conditions

$$\int_C \frac{f_C(t)t^j}{X_+(t)}\,\mathrm{d}t = 0, \qquad j = 0, \dots, -\operatorname{ind}_C J_C - 1.$$

When these conditions hold, the decaying solution is unique. These statements include the regularity and normalisation conditions that an unrestricted slogan about factorisation would omit. They do not assert a general matrix Riemann–Hilbert theorem.

Summary.

- For $h \in C_c^1(\mathbb{R})$, the normalised Cauchy integral has jump h and is the unique uniformly decaying additive solution continuous up to the axis.
- Matching continuous boundary values glue holomorphically. Uniform decay then permits Liouville's uniqueness argument.
- The chosen positive rational coefficient has index zero and an explicit product factorisation, giving the stated unique Wiener–Hopf pair.
- Under the positive-exponent time transform, positive-time support corresponds to the upper solution component. General index-dependent existence is cited.

The final lecture returns to differential equations and compares asymptotic solutions across a turning point.

References

M. J. Ablowitz and A. S. Fokas, *Complex Variables: Introduction and Applications*, second edition, Cambridge, 2003, §7.2, pp. 517–519, §7.3.1, pp. 527–533, and §7.4, pp. 570–576. The closed-contour coefficient convention is $\Phi_+ = g\Phi_- + f$. Our rational real-line equation and product factors are stated separately, so the source's quotient and index conventions are not silently interchanged.